

Week 2 Discussion

Covers: the axioms and calculus of probabilities (Week 1, Lecture 1.2) and counting (Week 2, Lecture 2.1).

Problem 1 (Working with the probability rules)

In a class, a student is chosen at random. Let R be the event that the student uses R and Y the event that the student uses Python, with

$$P(R) = 0.60, \quad P(Y) = 0.45, \quad P(R \cap Y) = 0.25.$$

For each answer below, name the rule you used. The relevant rules, for any events A and B :

- **Theorem 1.2.8 (complement rule):** $P(A^c) = 1 - P(A)$; also $P(\emptyset) = 0$ and $P(A) \leq 1$.
- **Theorem 1.2.9:**
 - **(a)** $P(B \cap A^c) = P(B) - P(A \cap B)$;
 - **(b)** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$;
 - **(c)** if $A \subset B$, then $P(A) \leq P(B)$ (monotonicity).
- **Bonferroni's inequality (1.2.9):** $P(A \cap B) \geq P(A) + P(B) - 1$.

(a) Find the probability that the student uses

1. at least one of the two languages;
2. neither language;
3. R but not Python;
4. exactly one of the two languages.

Hint: write each event in set notation first, for example “R but not Python” is $R \cap Y^c$.

(b) Now suppose you are told only $P(R) = 0.60$ and $P(Y) = 0.45$, but not $P(R \cap Y)$. What are the smallest and largest possible values of $P(R \cap Y)$? Could R and Y be disjoint?

Hint: one bound comes from Bonferroni's inequality, the other from monotonicity ($R \cap Y \subset Y$).

(c) Show that both bounds in (b) can actually be attained. Take a class of 100 students, each equally likely to be chosen, and describe which students use each language.

(a)

1. $P(R \cup Y) = P(R) + P(Y) - P(R \cap Y) = 0.60 + 0.45 - 0.25 = 0.80$ (Theorem 1.2.9(b)).
2. $P(R^c \cap Y^c) = P((R \cup Y)^c) = 1 - 0.80 = 0.20$ (DeMorgan, then the complement rule).
3. $P(R \cap Y^c) = P(R) - P(R \cap Y) = 0.60 - 0.25 = 0.35$ (Theorem 1.2.9(a)).
4. “Exactly one” is the disjoint union $(R \cap Y^c) \cup (R^c \cap Y)$, so its probability is $0.35 + (0.45 - 0.25) = 0.55$. Equivalently, $P(R \cup Y) - P(R \cap Y) = 0.80 - 0.25 = 0.55$.

(b) By Bonferroni, $P(R \cap Y) \geq 0.60 + 0.45 - 1 = 0.05$. By monotonicity, $P(R \cap Y) \leq P(Y) = 0.45$. So $0.05 \leq P(R \cap Y) \leq 0.45$. In particular $P(R \cap Y) > 0$, so R and Y cannot be disjoint.

(c) Number the students $1, \dots, 100$ and let $R = \{1, \dots, 60\}$.

- Lower bound: $Y = \{56, \dots, 100\}$ gives $R \cap Y = \{56, \dots, 60\}$, probability 0.05. Here every student uses at least one language.
- Upper bound: $Y = \{1, \dots, 45\} \subset R$ gives $R \cap Y = Y$, probability 0.45. Here every Python user also uses R.

Problem 2 (The matching problem)

Each of n letters is placed at random into one of n addressed envelopes, one letter per envelope, so all $n!$ arrangements are equally likely. What is the probability that **at least one** letter goes into its correct envelope?

Let A_i be the event that letter i is in its correct envelope. We want $P(\bigcup_{i=1}^n A_i)$. The events overlap, so we use the general **inclusion–exclusion** formula, which extends the two- and three-set versions from lecture:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P(A_1 \cap \cdots \cap A_n).$$

(a) Find $P(A_i)$. *Hint:* if letter i is fixed in its envelope, how many ways can the other $n - 1$ letters be arranged?

(b) Find $P(A_i \cap A_j)$ for $i \neq j$, and more generally $P(A_{i_1} \cap \cdots \cap A_{i_k})$ for any k distinct letters.

(c) How many terms are in the k th sum $\sum_{i_1 < \cdots < i_k}$? Show that the whole k th sum equals $1/k!$.

(d) Conclude that

$$P(\text{at least one match}) = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1} \frac{1}{n!}.$$

(e) Evaluate for $n = 3$ and $n = 4$. Check $n = 3$ by listing all $3! = 6$ arrangements. What happens as $n \rightarrow \infty$? *Hint:* recall $e^x = \sum_{k=0}^{\infty} x^k/k!$.

For $n = 3$: $1 - \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$. (Of the 6 arrangements, only the two cyclic ones, 231 and 312, have no match.) For $n = 4$: $1 - \frac{1}{2} + \frac{1}{6} - \frac{1}{24} = \frac{15}{24} = 0.625$.

As $n \rightarrow \infty$, the sum converges to $1 - e^{-1} \approx 0.632$. Surprisingly, the answer barely depends on n : with 10 letters or 10 million, the chance of at least one match is about 63%.

Problem 3 (Galileo's dice problem)

Around 1620, gamblers asked Galileo a puzzle: when three dice are rolled, a sum of **9** and a sum of **10** can each be made in six ways, yet experience showed that 10 comes up more often. Why?

(a) List the six unordered ways to write 9 as a sum of three dice values (for example $\{1, 2, 6\}$), and do the same for 10.

(b) The gamblers' reasoning treats the unordered outcomes as equally likely. Which sample space is equally likely for three fair dice, and how many outcomes does it have?

(c) For each unordered outcome in (a), count the ordered outcomes that produce it. *Hint*: use the multinomial coefficient. For example, $\{1, 4, 4\}$ corresponds to $\frac{3!}{1!2!} = 3$ ordered rolls.

(d) Compute $P(\text{sum} = 9)$ and $P(\text{sum} = 10)$.

(e) How many distinct *unordered* outcomes are there for three dice? Use the unordered-with-replacement formula from Table 1.2.1, and explain why this number cannot be used as $N(S)$ in $P(A) = N(A)/N(S)$.

Sum = 9	Orderings	Sum = 10	Orderings
$\{1, 2, 6\}$	6	$\{1, 3, 6\}$	6
$\{1, 3, 5\}$	6	$\{1, 4, 5\}$	6
$\{1, 4, 4\}$	3	$\{2, 2, 6\}$	3
$\{2, 2, 5\}$	3	$\{2, 3, 5\}$	6
$\{2, 3, 4\}$	6	$\{2, 4, 4\}$	3
$\{3, 3, 3\}$	1	$\{3, 3, 4\}$	3
Total	25	Total	27

With $6^3 = 216$ equally likely ordered outcomes, $P(\text{sum} = 9) = 25/216 \approx 0.116$ and $P(\text{sum} = 10) = 27/216 = 0.125$.

There are $\binom{6+3-1}{3} = \binom{8}{3} = 56$ unordered outcomes, but they are not equally likely: $\{3, 3, 3\}$ has probability $1/216$, while $\{1, 2, 6\}$ has probability $6/216$. This is Example 1.2.19 again: with replacement, count ordered outcomes.